

Preferences, Utility and Choice

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GR6211 - Microeconomic Analysis 1 - 2026

1 A Representation Theorem for Utility Maximization

Data

The Model

The Conditions

2 A Model Of Preference Maximization

3 Representation Theorems: Proofs

4 Uniqueness

A Representation Theorem

- When dealing with models that have latent (or unobservable) variables (such as utility maximization) we will want to find a *representation theorem*
- This consists of three things
 - A data set
 - A model
 - A set of conditions on the data which are **necessary** and **sufficient** for it to be consistent with the model
- A representation theorem tells us the observable implications of a model with unobservables
 - Means testing these conditions is the same as testing the model itself
- Often a representation theorem will have an associated **uniqueness result**
 - Tell us how precisely we have pinned down the unobservable variables

A Representation Theorem for Utility Maximization

- We are now going to develop a representation theorem for the model of utility maximization
- This is largely just formalizing the intuition we developed on the previous slides
- It is going to lead us to introduce a new model - that of preference maximization.

① A Representation Theorem for Utility Maximization

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- The data we are going to use are the choices people make
- Notation:
 - X : **Finite** set of objects you might get to choose from
 - 2^X : The power set of X (i.e. all the subsets of X)
 - $\emptyset$: The empty set
- Our data is going to take the form of a **choice correspondence** which tells us what the person chose from each subset of X

Definition

A choice correspondence C is a mapping $C : 2^X / \emptyset \rightarrow 2^X / \emptyset$ such that $C(A) \subset A$ for all $A \in 2^X / \emptyset$.

- This is just a way of recording what we described previously
- For example, if we offered someone the choice of Jaffa Cakes and Kit Kats, and they chose Jaffa Cakes, we would write

$$C(\{kitkat, jaffacakes\}) = \{jaffacakes\}$$

- C is just a record of the choices made from all possible choice sets
 - i.e. all sets in 2^X apart from the empty set $\emptyset$
- We insist that the DM chooses something that was actually in the data set
 - i.e. $C(A) \subset A$
- **Important:** Choice correspondence is non-empty: something is chosen from each choice set

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- ① X Finite
 - ② Observe choices from all choice sets
 - ③ We allow for people to choose more than one option!
 - i.e. we allow for data of the form

$$C(\{kitkat, jaffacakes, lays\}) = \{jaffacakes, kitkat\}$$

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- Which we interpret as something like “the decision maker would be happy with either jaffa cakes or lays from this choice set”

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- Which we interpret as something like “the decision maker would be happy with either jaffa cakes or lays from this choice set”
- These assumptions make our life easier, but are undesirable
 - We will relax them in later lectures

- Also, note that we are implicitly assuming that choice *only depends on the elements in A*
- Not (for example)
 - The order in which they are presented
 - A reference point
 - The amount of time people have to think
 - etc.
- We will come back to this when we discuss some of the evidence for and against utility maximization

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- Such that the things that are chosen are those which maximize utility
 - For every A

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$$C(A) = \arg \max_{x \in A} u(x)$$

- If this is true, we say that u **rationalizes** C
- If C can be rationalized by some u then we say it has a **utility representation**

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- We want to know when data is consistent with utility maximization
 - i.e. it has a utility representation
- So we would like to find a set of conditions on C such that it has a utility representation **if and only if** these conditions are satisfied
 - Testing these conditions is then the same as testing the model of utility maximization

- You may remember a condition called the Weak Axiom of Revealed Preference from Intermediate Micro

If $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B) \Rightarrow x \in C(B)$

Representation Theorem

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- You can (and will) show that α and β are equivalent to WARP
 - i.e. a data set satisfies α and β iff it satisfies WARP
 - α is 'from large to small'
 - β is 'from small to large'

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- You can (and will) show that α and β are equivalent to WARP
 - i.e. a data set satisfies α and β iff it satisfies WARP
 - α is 'from large to small'
 - β is 'from small to large'
- Notice we can **test** these conditions!
- If we have data, we can see if they are satisfied

- These conditions form the basis of our first representation theorem

Theorem

A Choice Correspondence on a finite X has a utility representation
if and only if *it satisfies axioms α and β*

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Theorem

A Choice Correspondence on a finite X has a utility representation
if and only if *it satisfies axioms α and β*

- **if:** if α and β are satisfied then a utility representation exists
- **only if:** if a utility representation exists then α and β are satisfied
- α and β are the exact empirical content of the model of utility maximization

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- We are going to prove this theorem
- Before we do so, we are going to introduce the notion of **preferences**, and the associated model of **preference maximization**
- Will explain why after we have introduced the model

- Consider the alternatives in X
 - e.g. Jaffa cakes, Kit kat, Lays
- Consider an exhaustive list of questions:

Do you like alternative x as much as alternative y ?

- If the answer is yes, then we write $x \succeq y$

The Questionnaire

Do you like...	Answer	We write...
j as much as j	yes	$j \succeq j$
k as much as k	yes	$k \succeq k$
l as much as l	yes	$l \succeq l$
j as much as k	yes	$j \succeq k$
k as much as j	no	
j as much as l	yes	$j \succeq l$
l as much as j	no	
k as much as l	yes	$k \succeq l$
l as much as k	no	

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 - Could extract them from choices (we will come back to this)
 - But we could ask people to express preferences over objects that we couldn't actually give them....

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 - Could extract them from choices (we will come back to this)
 - But we could ask people to express preferences over objects that we couldn't actually give them....
- Note that this is slightly different from the definition of questionnaire Q in Rubinstein's book
 - In fact it is his questionnaire R

- Technically speaking $\succeq$ is a **binary relation**

Definition

Consider a set X and denote by $X \times X$ its Cartesian Product. A binary relation B on X is a subset of $X \times X$. We write $B \subseteq X \times X$ and xBy if $(x, y) \in B$.

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- Example: for

	l	k	j
l	✓	✓	✓
k		✓	✓
j			✓

- is equivalent to

$$jBj, jBl, jBk, kBk, kBl, lBl$$

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- Examples of other binary relations
 - $X = \mathbb{R}$, $B = \geq$
 - $X = \text{set of political candidates}$, $B = \text{"would beat in a head to head contest"}$

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- Should we allow any possible answers to the questionnaire?
- No! Or at least we are going to rule some things out.
 - You cannot answer 'I don't know' or 'I like x much more than y ' (only yes or no answers)
 - You have to answer 'yes' at least one of the questions
 - Do you like alternative x as much as alternative y ?
 - or
 - Do you like alternative y as much as alternative x ?
 - Coherence
 - If you like x as much as y and y as much as z you must say that you like x as much as z

- Do these seem like sensible properties?
 - First, what do we mean by 'sensible'?
 - Normative vs Positive statements

- Do these seem like sensible properties?
 - First, what do we mean by 'sensible'?
 - Normative vs Positive statements
- Possible positive issues
 - Do you prefer 1 grain of sugar to 0 grains of sugar in your coffee?
 - Do you prefer a sun hat to a rain coat?
 - Do you prefer txuleta or oilasko for dinner?
 - Aggregation:

	A	B	C
1st	j	k	l
2nd	k	l	j
3rd	l	j	k

- Majority rule will lead to a violation of transitivity (a **Condorcet cycle**)

- These restrictions mean that the binary relation $\succeq$ has certain properties
 - Completeness: for every x and y in X either $x \succeq y$ or $y \succeq x$ (or both)
 - Transitivity: if $x \succeq y$ and $y \succeq z$ then $x \succeq z$
 - Reflexive : $x \succeq x$ for every x in X

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- There are many other properties one can define on binary relations, for example
 - Antisymmetric: $xRyRx$ implies $x = y$
 - Asymmetric: If xRy then not yRx
 - Symmetry: xRy implies yRx

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- There are many other properties one can define on binary relations, for example
 - Antisymmetric: $xRyRx$ implies $x = y$
 - Asymmetric: If xRy then not yRx
 - Symmetry: xRy implies yRx
- Are there circumstances in which $\succeq$ would have these properties?

- Let X be a non-empty set and R a binary relation on X

Definition

If R is transitive and reflexive then it is a **preorder**. If it is also antisymmetric it is a **partial order**. If it is also complete it is a **linear order**

Definition

(X, R) is a **preordered set** if R is a preorder, a **poset** if R is a partial order and a **loset** if R is a linear order

Definition

We will say R is a **preference relation** if it is a complete preorder

- Note that some people (mainly weird decision theorists) will use preference relation to refer to a preorder
- Note also that completeness implies reflexivity, so we can drop this from the definition of preference relation

- Notice that we can use $\succeq$ to define other binary relations:

- Strict Preference

$$x \succ y : \text{if } x \succeq y \text{ but not } y \succeq x$$

- This is called the **asymmetric** part of $\succeq$
- Indifference

$$x \sim y : \text{if } x \succeq y \text{ and } y \succeq x$$

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$$x \sim y : \text{if } x \succeq y \text{ and } y \succeq x$$

- This is called the **symmetric** part of $\succeq$
- What properties do these binary relations have?
 - Complete?
 - Transitive?
 - Asymmetric?
 - Symmetric?

- We can use preferences to form a model of choice
- We say that the complete preference relation $\succeq$ represents a choice function C if, for every A

$$C(A) = \{x \in A \mid x \succeq y \text{ for all } y \in A\}$$

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- i.e. the things that are chosen are those that are preferred to everything else in the choice set
- Note $\{x \in A \mid x \succeq y \ \forall y \in A\}$ are the $\succeq$ -maximal elements in A
 - If X is finite can we guarantee the existence of $\succeq$ -maximal elements?

- I hope you agree that the above concepts are well defined
- But why do we want to introduce the idea of preferences and preference maximization?

- I hope you agree that the above concepts are well defined
 - But why do we want to introduce the idea of preferences and preference maximization?
- ① Preference maximization is in some sense a more 'honest' model
 - Because they are ordinal provide a unique representation of choice, while utility does not
 - ② It is often convenient to treat preferences as data
 - Preferences may in fact be the primitive
 - Even if not, translation from choice to preference relatively straightforward
 - When dealing with more complex models of choice, it can be easier to start with the assumption of a well behaved preference relation, then add further conditions
 - Will see this when we talk about expected utility theory
 - ③ Introducing preferences will help us prove our representation theorem for utility maximization

Preferences, Utility, and Choice

- We are now going to use the concept of preferences to prove our representation theorem for utility
- In doing so we are going to link together choice, preferences, and utility
- We have already seen how we will link choice and preferences
- To link preferences and utility we can treat preferences as data and prove representation theorems of that type
- We say that a utility function u **represents** preferences $\succeq$ if

$$\begin{aligned} u(x) &\geq u(y) \text{ if and only if} \\ x &\succeq y \end{aligned}$$

Preferences, Utility, and Choice

- In fact, this is how we are going to prove our representation theorem
 - If we can find
 - A preference relation which represents choices
 - A utility function which represents preferences
- we are done!

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- If we can find
 - A preference relation which represents choices
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we are done!

- Preferences represents choices means

$$C(A) = \{x \in A | x \succeq y \text{ for all } y \in A\}$$

- Utility represents preferences means

$$u(x) \geq u(y) \iff x \succeq y$$

- So

$$\begin{aligned} C(A) &= \{x \in A | u(x) \geq u(y) \text{ for all } y \in A\} \\ &= \arg \max_{x \in A} u(x) \end{aligned}$$

- Thus, in order to prove that axioms α and β are equivalent to utility maximization we will do the following
 - ① Show that if the data satisfies α and β then we can find a preference relation $\succeq$ which represents the data
 - ② Show that if a binary relation is complete and transitive then we can find a utility function u which represents them
 - ③ Show that if the data has a utility representation then it must satisfy α and β (this you will do for homework)

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Theorem

Let C be a choice correspondence on a set X . Then there exists a preference relation $\succeq$ which represents C - i.e.

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Sufficiency: (Sketch - details in class):

- 1 Define candidate relation $\succeq$ using binary choice
- 2 Show that $\succeq$ is a preference relation
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Proof.

Necessity - Postponed for later



Theorem

Let $\succeq$ be a binary relation on a **finite** set X . Then there exists a utility function $u : X \rightarrow \mathbb{R}$ which represents $\succeq$: i.e.

$$\begin{array}{l} u(x) \geq u(y) \text{ if and only if} \\ x \succeq y \end{array}$$

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Proof.

We are going to use an indirect method that will help us later on

- 1 Proof by induction on the size of set $N = |X|$
- 2 Remove x^* , arbitrary element of X
- 3 By inductive assumption exists a utility function on X/x^*
- 4 Show that one can then assign utility to x^*

Proof By Induction - Reminder

- Here we have used **proof by induction**
 - We want to show that our statement is true regardless of the size of X
 - We do this using induction on the size of the set
 - Let $n = |X|$, the size of the set

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- Induction works in two stages
 - Show that the statement is true if $n = 1$
 - Show that, if it is true for n , it must also be true for any $n + 1$

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- This allows us to conclude that it is true for n
 - It is true for $n = 1$
 - If it is true for $n = 1$ it is true for $n = 2$
 - If it is true for $n = 2$, it is true for $n = 3...$

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 - It is true for $n = 1$
 - If it is true for $n = 1$ it is true for $n = 2$
 - If it is true for $n = 2$, it is true for $n = 3...$
- You have to be a bit careful with proof by induction
 - Or you can prove that all the horses in the world are the same color

- For homework you will show that if a choice correspondence has a utility representation then it must satisfy α and β
- Note that, with the proofs we have just done, this means that we have proved our main theorem

Theorem

A Choice Correspondence on a finite X has a utility representation
if and only if *it satisfies axioms α and β*

- Now we have proved this theorem let me provide some commentary

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① **IIA is not the same as rationality :**

- The following looks like a violation of α , but is it 'irrational'?

$C(\text{steak tatre, chicken, frogs legs}) = \text{steak tatre}$

$C(\text{steak tatre, chicken,}) = \text{chicken}$

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② Do not over interpret

- If someone's choices satisfy WARP, does this mean that they are maximizing utility?

③ What are the advantages of providing the representation theorem?

- Testability
- Providing an understanding of the model
- Allow us to compare different models more easily
- Question: Are all axioms testable?

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How Unique is Our Utility Function?

- We now know that if α and β are satisfied, we can find **some** utility function that will explain choices
- Is it the only one?

How Unique is Our Utility Function?

Croft's Choices	
Available Snacks	Chosen Snack
Jaffa Cakes, Kit Kat	Jaffa Cakes
Kit Kat, Lays	Kit Kat
Lays, Jaffa Cakes	Jaffa Cakes
Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- These choices could be explained by $u(J) = 3$, $u(K) = 2$, $u(L) = 1$

How Unique is Our Utility Function?

Croft's Choices	
Available Snacks	Chosen Snack
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Kit Kat, Lays	Kit Kat
Lays, Jaffa Cakes	Jaffa Cakes
Kit Kat, Jaffa Cakes, Lays	Jaffa Cakes

- These choices could be explained by $u(J) = 3$, $u(K) = 2$, $u(L) = 1$
- What about $u(J) = 100000$, $u(K) = -1$, $u(L) = -2$?

How Unique is Our Utility Function?

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Kit Kat, Lays	Kit Kat
Lays, Jaffa Cakes	Jaffa Cakes
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- These choices could be explained by $u(J) = 3$, $u(K) = 2$, $u(L) = 1$
- What about $u(J) = 100000$, $u(K) = -1$, $u(L) = -2$?
- Or $u(J) = 1$, $u(K) = 0.9999$, $u(L) = 0.998$?

How Unique is Our Utility Function?

- In fact, if a data set obeys α and β there will be **many** utility functions which will rationalize the data

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Theorem

Let $u : X \rightarrow \mathbb{R}$ be a utility representation for a Choice Correspondence C . Then $v : X \rightarrow \mathbb{R}$ will also represent C if and only if there is a strictly increasing function T such that

$$v(x) = T(u(x)) \quad \forall x \in X$$

- Any utility function that contains the same **ordinal** information will represent the same choices

How Unique is Our Utility Function?

Snack	u	v	w
Jaffa Cake	3	100	4
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- v is a strictly increasing transform on u , and so represents the same choices
- w is not, and so doesn't
 - For example think of the choice set $\{k, l\}$
 - u says they should choose kit cat
 - w says they should choose lays

Why Does This Matter?

- It is important that we know how much the data can tell us about utility
 - This is equivalent to figuring out identification in econometrics
 - How well does our data identify utility?
- For example, our results tell us that there **is** a point in designing a test to tell whether people maximize utility
- But there is **no** point in designing a test to see whether the utility of Kit Kats is **twice** that of Lays
 - Assuming α and β is satisfied, we can always find a utility function for which this is true
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 - Assuming α and β is satisfied, we can always find a utility function for which this is true
 - And another one for which this is false!
- We can use choices to help us determine that the utility of Kit Kats is higher than the utility of Lays
- But nothing in our data tells us **how much** higher is the utility of Kit Kats

- Question: what is the equivalent uniqueness statement for the model of preference maximization?